



Original article

Brainstem atrophy is associated with fatigue in myelin oligodendrocyte glycoprotein antibody-associated disease

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ABSTRACT

Background: Fatigue is a common symptom in myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD), but its underlying mechanisms are not well understood.

Objective: To investigate MRI volumetric changes associated with fatigue in MOGAD patients.

Methods: Forty-three MOGAD patients and 29 healthy controls (HC) were recruited from January 2020 to December 2022. Brain volumes, including the brainstem and its subregions (midbrain, superior cerebellar peduncle [SCP], pons, and medulla), were analyzed using FreeSurfer v7.1. Volumetric measures were compared between groups, adjusted for age, sex, and intracranial volume. Correlations between Fatigue Severity Scale (FSS) scores and brain volumes were examined, and predictors of fatigue severity were further explored using multiple linear regression analysis.

Results: Fatigue was present in 30.2 % of MOGAD patients. Fatigued patients had significantly reduced medulla volume compared to HC ($p = 0.004$) and exhibited higher Expanded Disability Status Scale scores, greater depression, and poorer sleep quality, and lower quality of life compared to non-fatigued patients. Higher fatigue scores correlated with smaller brainstem volumes, including the medulla ($r = -0.368$, $p = 0.019$), pons ($r = -0.331$, $p = 0.037$), and SCP ($r = -0.338$, $p = 0.033$). In multiple regression analysis, reduced medulla ($B = -0.014$, $p = 0.019$) and SCP volumes ($B = -0.188$, $p = 0.008$) were significant predictors of fatigue, while other clinical factors were not.

Conclusions: Fatigue in MOGAD is associated with reduced brainstem volumes, particularly in the medulla, suggesting a role of the brainstem in the pathophysiology of fatigue.

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1. Introduction

Chronic fatigue is common in central nervous system (CNS) autoimmune disorders, affecting up to 78–81 % of multiple sclerosis (MS) patients (Broch et al., 2021; Moore et al., 2022; Oliva et al., 2021) and 58–75 % of neuromyelitis optica spectrum disorder (NMOSD) patients. (Seok et al., 2022; Yeo et al., 2020) During immune system activation, inflammation can alter neuronal functions involved in arousal, emotional regulation, and motivation, contributing to fatigue. (Manjaly et al., 2019) This disabling symptom significantly reduces quality of life and poses a socioeconomic burden, including decreased work efficiency and potential job loss. (Oliva et al., 2021)

While research on fatigue in myelin oligodendrocyte glycoprotein antibody-associated disease (MOGAD) remains limited, a cross-sectional study suggested that approximately 75 % of patients may frequently and severely experience fatigue. (Ladakis et al., 2022) However, the anatomical and pathophysiological basis of fatigue in MOGAD remains unexplored. By contrast, MS and NMOSD have been more extensively studied, offering insights into possible structural and functional contributors to fatigue. In MS, thalamic and striatal atrophy, cortical thinning, and disrupted functional connectivity in networks such as the default mode and sensorimotor networks have been implicated. (ÁJ et al., 2013; Andreasen et al., 2010; Barbi et al., 2022; Bisecco et al., 2018; Palotai and Guttmann, 2020) Meanwhile, NMOSD-related fatigue has been linked to right thalamic atrophy and increased left pallidal volume, though cortical involvement appears minimal in this condition (Seok et al., 2022; Zhang et al., 2022)

Given the overlapping yet distinct neuropathological features of MOGAD, identifying structural correlates of fatigue in this disease is crucial. Prior studies have noted frequent lesion involvement in both supratentorial and infratentorial regions in MOGAD, suggesting that multiple structures—including cortical, subcortical, and brainstem regions—could contribute to fatigue. In particular, the thalamus has been implicated in fatigue in MS and NMOSD, (Palotai and Guttmann, 2020; Seok et al., 2022) while the brainstem serves as a key integrative hub for arousal, autonomic regulation, and sensorimotor processing. (Mohamed et al., 2024) Notably, brainstem lesions are prevalent in MOGAD. (Banks et al., 2021) Here, we employ volumetric MRI analyses across multiple brain regions to investigate structural correlates of fatigue severity in MOGAD.

2. Methods

We prospectively enrolled patients with MOGAD who visited the neurology outpatient clinic at the Samsung Medical Center in Seoul, South Korea between January 2020 and December 2022. The inclusion criteria were (1) being older than 18 years, (2) meeting the MOGAD diagnostic criteria, (Banwell et al., 2023) (3) experiencing at least 6 months of remission, and (4) lacking pre-diagnostic medical conditions that could contribute to fatigue, such as major depressive disorder, chronic infectious diseases, or chronic pain disorders. The patients underwent brain MRI and, within a 2-month period, also completed fatigue questionnaires of the Fatigue Severity Scale (FSS), as well as assessments using Beck's Depression Inventory (BDI-II), Pittsburgh Sleep Quality Index (PSQI), and Brief Pain Inventory (BPI). Two patients with MOGAD were excluded due to poor image quality on MRI. Baseline characteristics including age, sex, disease duration, number of attacks, the location of relapses, Expanded Disability Status Scale (EDSS) score, and current medications were collected. In addition, we recruited age- and sex-matched healthy controls (HC) with no prior medical, neurological, or psychiatric conditions. The study was approved by the Institutional Review Board of Samsung Medical Center, and written informed consent was obtained from all participants.

2.1. Questionnaires

The Korean versions of Fatigue Severity Score (FSS) was used for fatigue assessment (Ju et al., 2024) FSS comprises nine items evaluating fatigue levels on a scale from 0 to 7, resulting in a total score ranging from 0 to 63. Higher scores on the questionnaire indicate more severe fatigue. Patients with moderate-to-severe fatigue, defined as having a FSS score ≥ 36 , were classified as the fatigued group in this study. (Flachenecker et al., 2002)

The BDI-II is a 21-item self-rated questionnaire measuring the severity of depression, with scores ranging from 0 to 63, where higher scores indicate more severe depression. (Beck et al., 1996) A total score of 14–19 is classified as mild, 20–28 as moderate, and 29–63 as severe. The PSQI consists of 19 items evaluating subjective sleep quality over the past month. (Buysse et al., 1989) These items are combined into clinically relevant component scores, each weighted equally from 0 to 3. The total of the 7 component scores provides a global score ranging from 0 to 21, with higher scores indicating poorer sleep quality, and scores above 5 would be considered as a significant sleep disturbance. The BPI measures pain severity and its interference across 7 domains of functioning, using a scale of 0 to 10. We utilized the average pain severity score in this study. (Cleeland and Ryan, 1994)

2.2. MRI acquisition and image analysis

All participants underwent a three-dimensional volumetric brain MRI scan. An Achieva 3.0-Tesla MRI scanner (Philips, Best, the Netherlands) was used to acquire 3D T1 Turbo Field Echo MRI data using a sagittal slice thickness of 1.0 mm, over contiguous slices with 50 % overlap and no gap, a repetition time (TR) of 9.9 ms, an echo time (TE) of 4.6 ms, a flip angle of 8° and matrix size of 240 × 240 pixels reconstructed to 480 × 480 over a field of view of 240 mm. All axial sections were acquired parallel to the anterior commissure–posterior commissure line and perpendicular to the mid-sagittal plane.

All MRI data were processed using FreeSurfer version 7.1.1 (<https://surfer.nmr.mgh.harvard.edu>). FreeSurfer is a freely available software package providing automated brain segmentation, and neuroanatomical volume measurements. (FreeSurfer, 2012) We performed the standard processing pipeline following its recommended reconstruction procedure, including motion correction, skull stripping, and brain segmentation. Visual validation and manual correction were conducted to minimize errors in the automated processing. Additionally, FreeSurfer supports automated anatomical labeling of MRI data, providing whole brain volume, cortical and subcortical gray matter volume, cerebral white matter volume, and intracranial volume (ICV). In this study, we extracted four brainstem subregions (midbrain, pons, medulla oblongata, and superior cerebellar peduncle [SCP]) (Iglesias et al., 2015), and their volumes were estimated by their number of voxels.

2.3. Statistical analyses

Group comparisons of demographics and clinical features were conducted using the Chi-square test for categorical data and either the independent *t*-test or Mann-Whitney test for continuous data. Brain volumes were compared between HC and MOGAD patients using ANCOVA, adjusting for age, sex, and ICV. Bonferroni correction was applied for multiple comparisons. Correlations between brain volumes and clinical variables were assessed using Pearson's partial correlation analyses, controlling for age, sex, and ICV. Correlations between fatigue score (FSS) and other clinical variables were examined using Pearson or Spearman correlation, as appropriate. Multiple linear regression analysis was performed to identify predictors of fatigue severity (FSS), using demographic, clinical, and neuroimaging variables—including prior lesion locations and regional brain volumes—as independent variables. A *p*-value of <0.05 was considered statistically significant.

3. Results

The demographics, clinical characteristics and brain volume of the study participants are presented in Table 1. A total of 43 patients with MOGAD (22 females, 51.2 %; mean age, 43.7 ± 14.0 years) and 29 HC (16 females, 55.2 %; 39.0 ± 12.1 years) were included in the study. There were no significant differences in sex and age between the MOGAD and HC groups.

In MOGAD, the median disease duration was 3.4 (interquartile range [IQR], 1.5–5.9) years, and the median EDSS score was 1.5 (IQR, 1.0–2.5). The median number of total attacks was 2 (IQR, 1–3) and 65.2 % of patients experienced at least one relapse. Lesion locations were determined based on clinical and radiological findings across the entire disease course. Optic nerve was the most common attack location ($N = 33$, 76.7 %), followed by the cerebral hemisphere ($N = 12$, 27.9 %), brainstem/cerebellum ($N = 12$, 27.9 %), and spinal cord ($N = 8$, 18.6

Table 1
Demographics and clinical characteristics of study participants.

	MOGAD ($N = 43$)	MOGAD ($N = 43$)		<i>p</i> value
		Non-fatigued (FSS < 36, $N = 30$)	Fatigued (FSS ≥ 36, $N = 13$)	
Sex, female (%)	22 (51.2)	15 (50.0)	7 (53.8)	0.817
Age, years	43.7 ± 14.0	46.3 ± 13.8	37.9 ± 13.3	0.069
Disease duration, years	3.4 (1.5–5.9)	3.6 (1.6–6.1)	2.5 (1.0–4.8)	0.380
Total attack numbers	2 (1–3)	2 (1–3)	2 (1–3)	0.629
Annualized relapse rate	0.61 (0.42–1.35)	0.53 (0.36–1.32)	0.84 (0.48–1.85)	0.232
Relapsing course (%)	28 (65.2)	20 (66.7)	8 (61.5)	0.742
Attack sites (%) ^a				
Cerebral hemisphere	12 (27.9)	9 (30.0)	3 (23.1)	0.727
Brainstem/cerebellum	12 (27.9)	8 (26.7)	4 (30.8)	1.000
Spinal cord	8 (18.6)	5 (16.7)	3 (23.1)	0.681
Optic nerve	33 (76.7)	23 (76.7)	10 (76.9)	1.000
Presence of brainstem lesions ^b	10 (23.3)	7 (23.3)	3 (23.1)	1.000
EDSS	1.5 (1.0–2.5)	1.5 (0.5–2.0)	2.0 (1.5–3.375)	0.042
Fatigue severity scale	28.1 ± 15.1	19.8 ± 8.3	47.2 ± 7.9	<0.001
BDI-II	15.3 ± 11.9	10.3 ± 8.7	27.3 ± 10.1	<0.001
PSQI	6.5 ± 3.3	5.6 ± 2.8	8.6 ± 3.5	0.007
BPI	0.125 (0–2.5)	0 (0–2)	2 (0–3)	0.068
SF36				
PCS	67.2 ± 20.0	74.1 ± 16.1	51.1 ± 19.5	0.001
MCS	65.5 ± 18.1	71.4 ± 14.6	51.6 ± 18.6	0.001
Medications (%)				
steroids ^c	10 (23.3)	5 (16.7)	5 (38.5)	0.140
Other IS ^d	28 (65.1)	21 (70.0)	7 (53.8)	0.324

MOGAD, myelin oligodendrocyte glycoprotein antibody-associated disease; HC, healthy controls; EDSS, expanded disability status scale; BDI-II, Beck depression inventory-II; PSQI, Pittsburgh sleep quality index; BPI, brief pain inventory; SF36, short form-36 health survey; PCS, physical component summary; MCS, mental component summary; IS, immunosuppressants.

Continuous variables are expressed as mean $\pm$ standard deviation or median (interquartile range).

^a Duplicates are allowed.

^b Abnormal signal intensities or atrophic changes on T2- or T1-weighted MRI at the time of study inclusion.

^c methylprednisolone dose, median (interquartile range) = 10 (8.75–15) mg.

^d azathioprine (Imuran), mycophenolate mofetil (CellCept), methotrexate (Methotrexate).

%). Among patients with brainstem relapse, 25 % ($N = 3$) had widespread T2-hyperintense lesions spanning contiguous brainstem regions (midbrain, pons, and medulla) during the acute attack. The remaining patients had more localized lesions involving the midbrain and pons ($N = 2$), pons and medulla ($N = 1$), or a single region, including the midbrain ($N = 1$), pons ($N = 2$), or medulla ($N = 3$). At the time of MRI analysis, the majority of patients with prior brainstem relapses (10 out of 12, 83.3 %) exhibited abnormal T2 signal intensities or atrophic changes in the brainstem. Most spinal cord lesions were in the cervical spinal cord ($N = 7/8$, 87.5 %), with two involving the cervicomedullary junction.

The mean FSS score was 28.1 ± 15.1 , with 13 patients (30.2 %) experiencing moderate to severe fatigue, indicated by scores of 36 or higher. Depression at or above the moderate level was observed in 15 patients (34.9 %), including five with severe depression. The mean PSQI score was 6.5 ± 3.3 , and 21 patients (48.8 %) had poor sleep quality. The median average pain severity score among MOGAD patients was 0.125 (IQR, 0–2.5). Steroids were prescribed in 23 % of patients (mean dose; 10 ± 3.5 mg), and 65 % received additional immunosuppressive therapy, either with or without steroids.

When comparing MOGAD patients and HC, the volume of the medulla ($p = 0.031$) was significantly reduced in MOGAD patients. Cortical volume showed a trend toward reduction, with borderline significance ($p = 0.052$). No significant differences were found in the volumes of white matter, deep gray matter, cerebellum, whole brainstem, midbrain, SCP, or pons between MOGAD and HC.

3.1. MOGAD patients with fatigue

Fatigued MOGAD patients had higher EDSS scores, greater depressive symptoms, poorer sleep quality, and lower quality of life compared to non-fatigued patients (Table 1). In contrast, demographic and clinical characteristics including disease duration, total number of attacks, relapsing course, and current medication use, did not differ significantly between the two groups. The distribution of prior relapse locations (cerebral hemispheres, brainstem, spinal cord, and optic nerve) was also similar. Despite this comparable relapse pattern, only the fatigued MOGAD patients showed a significantly reduced medulla volume compared to HC ($p = 0.004$; Fig. 1), whereas the non-fatigued MOGAD patients did not differ significantly from HC.

Across the entire MOGAD cohort, higher fatigue scores were associated with reduced volumes of the medulla ($r = -0.368$, $p = 0.019$), pons ($r = -0.331$, $p = 0.037$), SCP ($r = -0.338$, $p = 0.033$), and whole brainstem ($r = -0.365$, $p = 0.021$) (Fig. 2). However, FSS scores were not significantly correlated with cortical and deep gray matter volumes, cerebral white matter volume, midbrain volume, or cerebellar volume. Regarding clinical variables, higher fatigue scores correlated with higher EDSS ($\rho = 0.349$, $p = 0.025$), PSQI ($\rho = 0.554$, $p < 0.001$), and BDI-II ($\rho = 0.630$, $p < 0.001$) scores, as well as lower PCS ($\rho = -0.581$, $p < 0.001$) and MCS ($r = -0.706$, $p < 0.001$) scores. No significant correlations were observed between

FSS scores and age, disease duration, relapsing course, total number of attacks, or current use of medications.

Among the EDSS, PSQI, and BDI-II scores, and the volumes of the medulla, pons, and SCP, a significant correlation was found only between BDI-II score and pons volume ($r = -0.371$, $p = 0.040$).

3.2. Predictors of fatigue severity (FSS) in mogad

In the multiple regression analysis, reduced volumes of medulla ($B = -0.014$, $p = 0.019$) and SCP ($B = -0.188$, $p = 0.008$) were significant predictors of increased fatigue severity (Table 2). In contrast, other clinical variables, including age, sex, disease duration, EDSS, PSQI, BDI-II scores, and current use of steroids or immunosuppressants, were not significantly associated with fatigue severity.

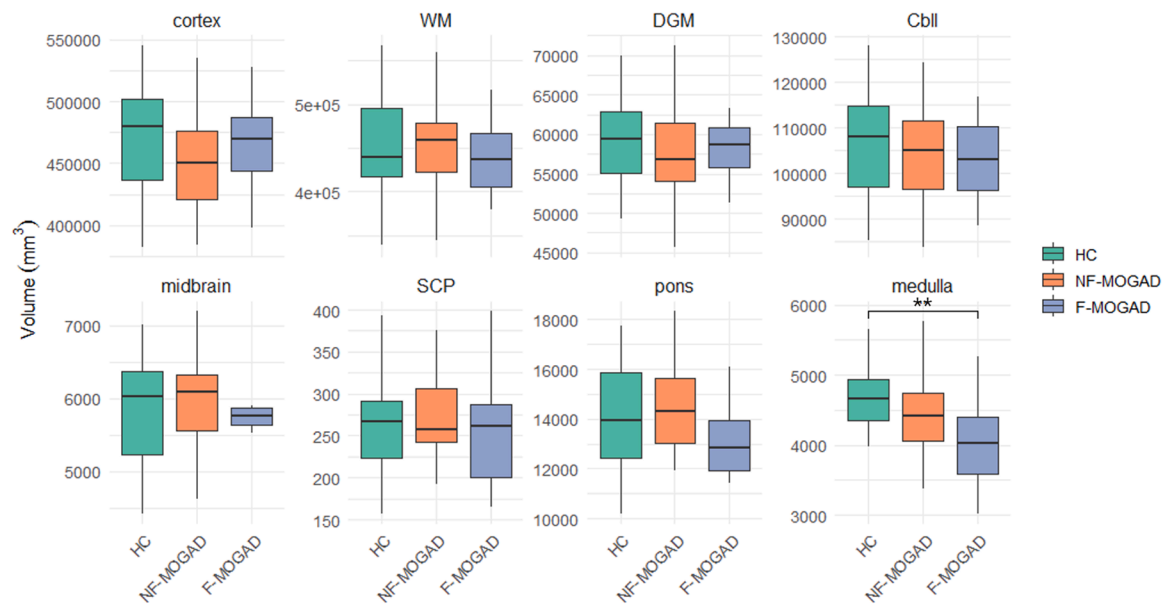


Fig. 1. Brain volume comparison across groups. Comparison of brain volume between groups using ANCOVA, controlling for age, sex, and intracranial volume: healthy controls (HC) vs MOGAD patients without fatigue (NF-MOGAD) vs MOGAD patients with fatigue (F-MOGAD). ** $p < 0.01$; if not indicated, no significant difference was detected. WM, cerebral white matter; DGM, deep gray matter; Cbll, cerebellum; SCP, superior cerebellar peduncle.

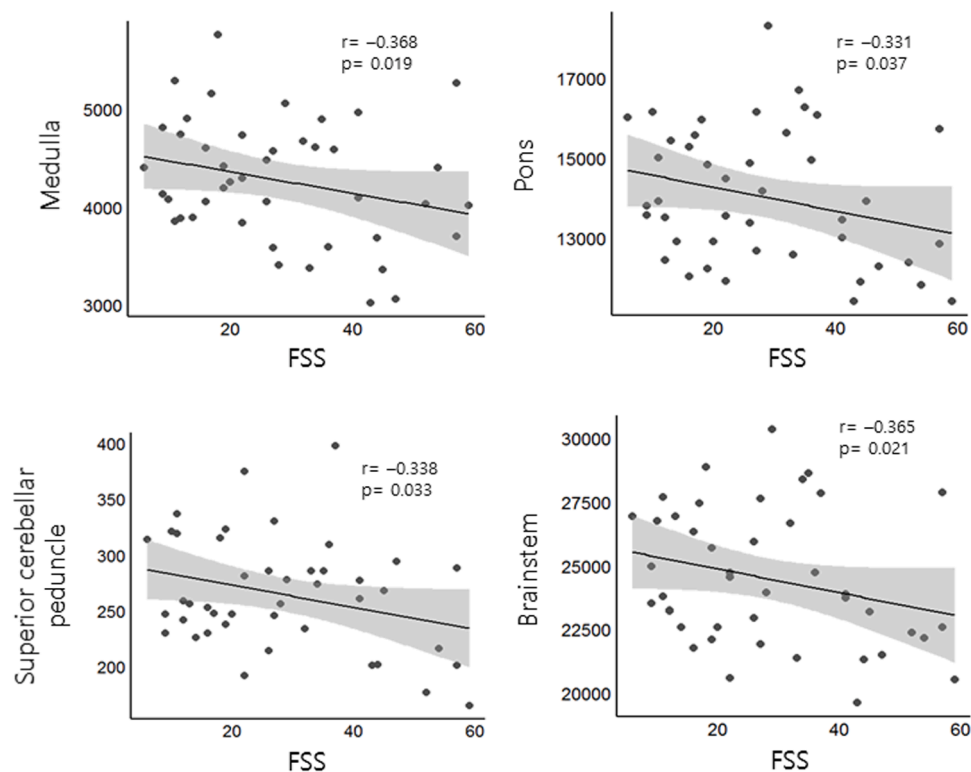


Fig. 2. Partial Pearson correlations between FSS scores and brainstem structure volumes. Scatter plots of partial Pearson correlations with 95 % confidence intervals between FSS scores and the volumes of the medulla, pons, superior cerebellar peduncle, and whole brainstem, all of which showed significant associations. FSS, fatigue severity scale.

4. Discussion

This single-center study provides novel evidence associating fatigue in MOGAD patients with structural atrophy in the brainstem, particularly in the medulla. Approximately one-third (30 %) of MOGAD patients experience clinically significant fatigue, with reduced medulla

volume being more pronounced in those reporting fatigue compared to HC. Fatigue levels were further influenced by factors such as depression, poor sleep quality, and severe disability. These findings emphasize the critical role of brainstem structural integrity in MOGAD-related fatigue.

Fatigue, a complex symptom seen in various CNS disorders, is often associated with changes in infratentorial structures. For example, in

Table 2
Predictors of fatigue severity in MOGAD.

	FSS ($R^2 = 0.654$)		
	B	SE	p value
Female	−5.378	5.157	0.316
Age	0.069	0.201	0.736
Disease duration	−0.189	0.674	0.783
Attack number	1.292	1.308	0.342
Relapsing course, yes	−4.139	5.486	0.464
Previous attack sites			
cerebral hemisphere	−12.506	8.173	0.150
brainstem/cerebellum	−1.567	5.416	0.777
spinal cord	10.923	5.247	0.058
optic nerve	−13.754	8.004	0.109
EDSS	−0.048	2.180	0.983
PSQI	1.402	1.056	0.207
BDI-II	0.134	0.297	0.660
BPI	1.935	2.095	0.373
Steroid use, yes	0.761	7.282	0.918
Other IS use, yes	−3.510	4.630	0.462
Volume (mm ³)			
midbrain	0.014	0.007	0.068
SCP	−0.188	0.060	0.008
pons	−0.001	0.002	0.509
medulla	−0.014	0.005	0.019

MOGAD, myelin oligodendrocyte glycoprotein antibody-associated disease; FSS, fatigue severity scale; EDSS, expanded disability status scale; BDI-II, Beck depression inventory-II; PSQI, Pittsburgh sleep quality index; BPI, brief pain inventory; IS, immunosuppressants; SCP, superior cerebellar peduncle; B, unstandardized beta coefficient; SE, standard error.

ME/CFS, significant microstructural abnormalities, particularly in the medulla, pons, and midbrain, have been identified, with increased transverse diffusivity in the medulla correlating with fatigue severity. (Thapaliya et al., 2021) Similarly, reduced white matter volume in regions such as the pons and midbrain has been noted. (Finkelmeyer et al., 2018) In post-coronavirus disease-2019 syndrome, microstructural changes in the brainstem regions, including the inferior medullary reticular formation and the raphe nuclei within the medulla, have been associated with disease severity, impaired functional recovery, and increased fatigue. (Rua et al., 2024) In MS, the role of the brainstem in fatigue has been less extensively studied; however, existing research points to a strong association between fatigue severity, infratentorial lesion volume and EDSS scores. (Ezzeldin et al., 2023) Longitudinal cohort studies in MS also suggest that pons volume could serve as a marker for fatigue progression. (Fleischer et al., 2022) Widespread white matter damage, including microstructural white matter abnormalities in the brainstem and cerebellum, has also been implicated in MS-related fatigue. (Novo et al., 2018) Collectively, these studies emphasize the importance of brainstem integrity in contributing to fatigue across various CNS disorders. Disruptions in monoaminergic neurotransmitter systems originating from dopaminergic, serotonergic, or noradrenergic nuclei, may impair mood, motivation, and arousal. (Carotenuto et al., 2023) Additionally, structural damage to the brainstem or hypothalamus could affect the hypothalamic-pituitary-adrenal axis or autonomic pathways, leading to prolonged energy deficits and physiological imbalances. (Manjaly et al., 2019)

In the context of MOGAD, the brainstem may play an even more prominent role in fatigue development. Diffuse lesions in the brainstem or cerebellar peduncles, often near the fourth ventricle, are more commonly observed in MOGAD compared to MS. (Banks et al., 2021) One study reported that 31.5 % of MOGAD patients showed brainstem or combined brainstem-cerebellar involvement during the disease course, with 76.1 % of these cases presenting at disease onset. (Xu et al., 2023) Consistent with this, our MR volumetric analysis revealed significant reductions in medulla volume, which closely correlated with fatigue severity. However, the literature presents varying results regarding brainstem volume changes in MOGAD; while one study have reported

brainstem atrophy as early as the first year post-diagnosis, (Rechtman et al., 2022) others found no significant differences between MOGAD patients and HC, with median disease durations ranging from 0.8 to 2 years. (Duan et al., 2021; Messina et al., 2022) Notably, no previous studies have specifically examined brainstem subregions, let alone their association with fatigue in MOGAD, highlighting the need for further research. Additionally, our study did not observe changes in gray matter, white matter, or deep gray matter volumes, which contrasts with findings from previous studies. (Duan et al., 2021; Kogel et al., 2024; Messina et al., 2022) Small sample sizes, differing methodologies, ethnic variations and heterogeneous clinical presentations may all contribute to these discrepancies.

It is important to note several limitations in this study. First, its cross-sectional design causal inferences between brainstem atrophy and fatigue. Second, the relatively small sample size and inclusion of participants from a single ethnicity may limit the generalizability of our findings. Additionally, the proportion of fatigued patients (30 %) in this study was lower than that reported in previous studies (72–75 %). (Kogel et al., 2024; Ladakis et al., 2022) This discrepancy may be attributed to differences in study design, or clinical heterogeneity among MOGAD patients. Third, we did not include functional imaging data or direct measures of autonomic dysfunction, which could provide further insights into the relationship between fatigue and brainstem atrophy. Finally, potential confounding factors, such as concurrent mood or sleep disorders and medication use, could have influenced fatigue severity and may also be associated with brainstem volume. However, it is notable that medulla volume in this study did not show a direct association with depression or sleep disturbances.

In conclusion, this study reveals a strong association between fatigue in MOGAD and structural atrophy in the medulla. The findings suggest the crucial role of brainstem integrity in the pathophysiology of fatigue. However, further research is necessary to identify the factors driving brainstem atrophy and to clarify the mechanisms connecting brainstem changes to fatigue in MOGAD.

Ethical considerations and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Institutional Review Board of Samsung Medical Center, and written informed consent was obtained from all participants.

Data availability

The dataset is available from the corresponding author upon reasonable request.

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CRedit authorship contribution statement

Eun Bin Cho: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Youjin Seo:** Software, Methodology, Formal analysis. **Daegyeom Kim:** Writing – review & editing, Formal analysis. **Seung Ho Choo:** Investigation, Data curation. **Yeon Hak Chung:** Methodology, Data curation. **Sung Tae Kim:** Software,

Data curation. **Cheol E. Han:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Ju-Hong Min:** Writing – review & editing, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

J-HM has lectured, consulted, and received Honoraria from Bayer Schering Pharma, Merck Serono, Biogen Idec, Sanofi Genzyme, Teva-Handok, UCB, Samsung Bioepis, Mitsubishi Tanabe Pharma and Roche; received a grant from the National Research Foundation of Korea and SMC Research and Development Grant. All other authors have no relevant financial or non-financial interests to disclose.

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